FishAI v1.5: A Bit-Budget Memory Ladder as an Honest Difficulty Axis

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Abstract

Until now the FishAI engine's easy tier was a six-event memory window plus a 25% uniform blunder rate — a difficulty knob that is neither monotone, interpretable, nor human-shaped. FishAI v1.5 replaces it with the bounded-memory model of Sanjay Kannan's Literature framework: memory capacity capped *in bits* under a fixed fact tariff (2 bits per card fact, 1 per basis fact), with the eviction policy the strategy corpus recommends — rank facts by contestability, not recency — implemented as a stateless, pure re-derivation from the full public log at every decision. The design is measured under the project's pre-registration discipline, eight predictions with verdict rules fixed before each run. The ladder is confirmed monotone across all ten measured budgets (P1: every adjacent rung non-negative, the sharpest at $z = 46.8$; P2: the ∞ cell an exact mirror over 3,000 duplicate pairs). Calibrating the shipped tiers onto the ladder (P3, mixed): medium is worth 32.2 bits $[30.8, 35.1]$; the old noise-based easy tier measures *below the zero-bit floor* (0.0446 set-share against the floor's 0.1313 — a memoryless bot that reasons beats a remembering bot that dices); hard sits above every finite rung. Evidence-age analysis (P4–P6, all mixed as printed) finds the S45 signature where the model predicts it — budgets up to 48 bits decay, with half-lives 5/13/17/33/49 events, rising in bits where defined — while the noise tier is cliff-edged at its window ($z = 258.9$) yet not flat inside it, and a composition effect makes the pooled and seed-clustered estimators disagree in sign for the full-memory policies. The headline refutation is P7: memory pressure was predicted to erode the behavioural style signature, and instead the v1.0 classifier reads bounded ecologies *better* — top-1 rises from 15.4% at 16 bits to 23.0% at 64 bits against 22.4% at ∞, and the violating rung (-0.0053 ± 0.0019) survives a Bonferroni $\times 3$ annotation at $p = 0.00885$. A single-seat follow-up (E4b), registered after review, isolates the read seat: its 50-seed pilot confirmed a flat curve but at 52% power — its top rung's confidence interval contained the entire P7 effect — and the pilot's commit overclaimed; the correction is on the record, and the 300-seed run of record (99.97% post-hoc power at the P7 effect size) confirms P8 flat at matched read count. Within the designs measured, the P7 shift is a property of the bounded ecology, not of the bounded seat's own end-of-game signature; the cross-design difference ($z = 2.11$) is reported as exactly that and enters no verdict. An addendum re-measures the arm after the engine gained a defusal term that all nine roster styles carry and v1.5 inherits: against the same pre-concession v1.0 whose committed cell reads -0.1255 ± 0.0590, the updated arm measures $+1.4065 \pm 0.1399$ and $+1.5290 \pm 0.1418$ on two banks — a different question, not a correction of that cell, which a per-pair audit (0/2,700 mismatches against a pre-concession export, 1,369/2,700 against the current tree) establishes was measured clean. The ladder's *shape* does not survive either: the $0 \to \infty$ span grows from 8.0195 to 9.7985 and 9.9260. And because the term's licence scan reads the full public log while retiring only against budgeted knowledge, it partly bypasses the bit budget — held against its own pre-concession self at a zero budget, the updated arm still takes +0.96 and +0.80 sets per pair on the same two banks — so "updated v1.5 at N bits" is not a clean N-bit agent, and the new ladder is not a pure memory ladder. The original one, measured before the term existed, is unaffected.

1 Introduction

Every difficulty system makes a claim about what a weaker player *is*. The FishAI engine's shipped answer, inherited from v0.5 [4], was the standard one: an easy bot remembers only the last six public events, skips constraint inference, and replaces 25% of its asks with random legal ones. That answer is expedient and wrong in three measurable ways. It is not monotone — noise interacts with position, so "more noise" is not reliably "weaker" in any calibrated sense. It is not interpretable — there is no single number a player can reason about. And it is not human-shaped: humans do not play well and then dice-roll into blunders; they *forget*, oldest and least relevant facts first, and their errors inherit that structure [9, 2].

The project's research synthesis catalogues an alternative with all three properties [9]: the bounded-memory player of Sanjay Kannan's Literature framework (S44 in the synthesis's numbering), in which an agent's memory is capped in *bits* under a fixed tariff per atomic fact, and the interesting decision — given k bits, what should you remember? — is delegated to an explicit eviction policy. The synthesis pairs it with two neighbours that complete the design: the recency player (S45), whose observable signature is a decay of effectiveness with the age of the supporting evidence, and the triage memoriser (S47), the human-facing name for the eviction policy the strategy corpus actually recommends — memorise the books you can contest, and flush what is resolved.

FishAI v1.5 implements that model as a first-class policy arm of the deterministic `us54` engine (Section 5), and then measures it under the discipline the project's adaptive suite established [5]: pre-registered predictions with verdict rules fixed before each run, duplicate deals, paired and seed-clustered standard errors, health gates that void a run rather than qualify it, and one committed artifact [17] from which every number in this paper is read. Four experiments and two registered follow-ups answer four questions. Is the ladder monotone? (Yes, at every rung.) Where do the shipped tiers land on it? (Medium at 32.2 bits; the old easy tier below the ladder's own floor.) Are bounded bots human-shaped in the S45 sense? (Where the model has room to bind, yes, with half-lives that rise in bits; the full-memory verdicts are mixed for reasons worth stating exactly.) And does memory pressure erode a seat's behavioural style signature, as the v1.0 observation work [6] would suggest? Here the pre-registered prediction was **refuted**, and the refutation is this paper's headline: under mild memory pressure, 64-bit ecologies are *more* classifiable than full-strength ones, and a registered single-seat follow-up — including a pilot whose conclusion had to be corrected on the record, a sequence Section 6.4 reports in full — localises the effect in the ecology rather than in the read seat's own signature, within the designs measured.

2 Related work

The primary source is Kannan's Literature framework and its reference implementation, both read in full by the project's research synthesis [3, 9]: the framework specifies an agent "constrained by imperfect memory" via a bit parameter and a per-fact price list, and its reference `ActivePlayer` implements ranked retention with a `spotlight()` focus-book mechanism — the direct ancestors of Section 5. The idea that agents are resource-bounded reasoners rather than noisy optimal ones goes back to Simon [1]; the specific empirical shape assumed here — retrieval success decaying with the age of the supporting evidence, at rates that track the environment — is the memory literature's [2], and the S45 entry of the synthesis [9] argues the decay curve is the single best skill estimator recoverable from a Literature log. Within the project, this paper leans on the v0.5 roster and measurement discipline [4], the v1.0 classifier whose accuracy harness E4 reuses [5, 6, 15], and the inert-axis result that bounds what any public-log observer can see of this roster [7, 11].

3 Background

Literature (Canadian Fish) under the `us54` dialect [10, 4]: 54 cards in nine half-suits ("books") of six, six players in two teams, public asks gated by a holding licence, out-of-turn declares in a deterministic window, any declaration error awarded to the opponents, race to five sets, ties impossible. The engine is pure, deterministic TypeScript; one seed reproduces a byte-identical game [19].

Three prior measurements matter here. First, the v0.5 roster: nine parameterised styles over one shared full-strength inference engine, with Balanced the tuned control [4]. Second, the difficulty tiers as shipped: `easy` is a six-event log window without set constraints, a 25% uniform ask-blunder rate, and no claim planner; `medium` runs the full knowledge engine; `hard` adds the constraint-refined hit probability — the skill at which every roster style, and the bounded arm below, plays [19]. Third, the v1.0 observation layer: fourteen public-log features, a calibrated diagonal-Gaussian classifier, and a measured end-of-game top-1 of 22.4% over nine styles against an 11.1% chance floor [6, 16]. Experiment E4 runs that exact instrument over bounded play.

4 The S44 model and the cost tariff

The model prices memory, not behaviour. An agent holds at most k bits of derived facts, where the tariff — fixed by the framework, not tuned by this project [3, 9] — is:

Fact kind	Reading	Cost
has-card	player X has card Z	2 bits
lacks-card	player X does not have card Z	2 bits
basis	player X holds at least one card of book Y	1 bit
no-basis	player X holds no card of book Y	1 bit

The framework's specification names the first three prices; its reference implementation's fact tuples carry the negative basis fact as a fourth kind, priced like the positive [3, 9]. Two properties of the tariff do real work. A basis fact costs half a card fact and is often worth more — it is what gates ask legality and drives hit priors — so a good eviction policy is not a compression problem but a valuation problem. And the framework does no enforcement: the agent gets `self.limit` bits, tracks `self.free`, and the discipline is its own; the reference `ActivePlayer` *refills memory from scratch each turn*, re-deriving every fact and re-spending the budget down a relevance ranking anchored on a focus book chosen by its `spotlight()` routine. That refill semantics — not an incremental store with evictions — is what Section 5 implements faithfully.

The synthesis's calibration argument [9] sets the scale: worst-case perfect recall of card facts alone runs to 2 bits per card per seat — hundreds of bits (the framework's own constants are for its 52-card, seven-card-major deck; the model transfers to `us54`, the constants do not). A two-digit budget is therefore a genuine handicap, and Section 7.1 measures that on these deals the axis saturates by 128 bits.

Why this replaces noise as the difficulty knob is the synthesis's argument, which the experiments test rather than assume: a bit budget is *monotone* (more bits can only add facts; P1), *interpretable* (one number a player can reason about), *human-shaped* (a relevance-ranked bounded memory forgets the oldest, least-contestable facts, so the S45 decay curve arises by construction; P5), and *composable* (a bounded Hoarder is still a Hoarder — the budget caps the memory, never the style or the reasoning; E4 exercises exactly this).

5 The engine: a stateless bounded-memory arm

The v1.5 arm is a fifth `PolicySpec` shape, `{bounded: true, bits, style}` — budget in bits, roster style to play over the restricted knowledge (default Balanced) — resolved inside `decide` exactly as the adaptive arm is [5, 19]. Everything below is a pure function of (view, policy, seed): no `Date`, no `Math.random`, no state between calls. The engine's own summary of the division of labour: the budget caps the memory, never the reasoning — the chosen roster style runs at hard skill over whatever knowledge survives the budget [19].

5.1 Derivation, then reconstruction

The implementation is two passes over public information [19]. **Derivation**: the full-fidelity recorded walk of the knowledge engine — the identical ingestion `buildKnowledge` runs at hard skill — read back as timestamped atomic facts of the four tariff kinds. Nothing is invented: a hit locates a card; an ask certifies the asker's basis and their lack of the named card; a miss certifies the target's lack; historical count exhaustion certifies eliminations; a claim resolves its book and retires every fact on it — exactly the certifications the observation layer documents for this log [6]. **Reconstruction**: the *kept* facts are replayed in log order onto a fresh working state through the same primitives the knowledge engine uses, then finished by the shared `finishKnowledge` step — own-hand injection, fixpoint propagation over the *current* public counts, materialisation. The output is the same `Knowledge` shape every style policy consumes, so all nine roster styles run over it unchanged.

Because the reconstruction reuses the knowledge engine's own machinery, the **large-budget equivalence anchor** holds by construction rather than by a parallel implementation staying in step: with every fact kept, the replay reproduces the full walk's state, and the arm's decisions are identical to the unbounded style's over whole games — pinned by test, and re-verified at run scale by the ∞ cells of Sections 7.1 and 7.5 [19, 17].

5.2 What is free, and what never enters the pool

Own hand is free — the seat can see it, so the injection runs at every budget, including zero. The public board state is free — resolved books are seeded gone from the view's book state, and current counts drive the final propagation — because the synthesis's S48 entry keeps *observation history* and *display policy* as separate axes: what the table shows is not what the agent remembers [9]. The S45 subtlety identified for the old log-window tier applies with force here: a fact-filtered replay of running hand counts would be wrong the moment a hit's fact was evicted, so the reconstruction never replays counts at all — count exhaustion runs only inside the full-fidelity derivation, and its conclusions reach the replay as kept lacks-card facts [9, 19].

Facts the free state reproduces at any budget are never enumerated, so the budget is never spent on them: everything on a resolved book; the viewer's own lacks and held cards; the viewer's own basis constraints. Subsumption keeps the accounting honest — a card fact subsumes the eliminations on its card, and a no-basis fact compresses up to twelve bits of per-card lacks into one [19].

5.3 Spotlight ranking and the budget

The eviction policy is S47's, under the reference implementation's mechanism: each live book is scored by how much the seat's own hand plus the derived facts bear on it (2 per held card, plus the bit cost of every pool fact on the book); facts rank by book score descending, then recency descending, then a stable total order (card in deck order, seat, kind); the kept set is the longest ranked prefix whose total cost fits the budget [19, 9]. Contestability leads and recency tiebreaks —

the corpus's advice, not this project's invention: save your memory for the half-suits you can still contest, and flush what is resolved [9]. At `bits: 0` nothing is kept and the policy sees own hand plus public board only — the synthesis's own-hand-only regime (S46), and still never an illegal move; the test suite pins the zero-budget floor, the budget accounting (kept cost never exceeds bits; the kept set is a ranking prefix; ties deterministic), `decide`/`decideExplained` parity for the new arm, and purity [19]. The explanation trace names the budget ("k-bit memory"), so downstream surfaces cannot misrepresent what played.

6 Experimental design and pre-registration

6.1 Shared discipline

All experiments inherit the adaptive suite's discipline [5, 12]: duplicate deals (each seeded deal played in both orientations, start seat rotating with the pair index), per-seed paired estimators for adjacent comparisons, seed-clustered standard errors wherever compositions share seeds, and health gates — 0 illegal actions, 0 capped games, 0 invariant violations, and under `us54` 0 ties, 0 voids, 0 non-clinch endings, with `distinctSeeds` equal to the expected count in every cell — that *void* a run rather than footnote it. Decisions are seeded `hashSeed(`${seed}:${moveIndex}`)()`; the runner is a pure function of its task (the worker pool is a throughput device, not a source of variation); an FNV digest of the records travels in the artifact; and the committed artifact `src/lab/data/bounded-results.json` sits behind a boundary validator mirroring the adaptive one [17, 19]. The base suite is 85,200 games in 308 seconds of wall clock; the E4b pilot and power runs add 7,200 and 43,200 games [17]. Every number in this paper is a field of that artifact (or of the committed v1.0 artifact where the baseline is cited [16]), quoted at the artifact's own precision.

6.2 The experiments

1. **E1 — ladder monotonicity.** A team of three bounded-Balanced seats against an unbounded Balanced team at hard skill — the same full-strength engine the budget caps — at ten budgets, bits $\in \{0, 8, 16, 24, 32, 48, 64, 96, 128, \infty\}$ (∞ implemented as 10^6, provably above the derivable pool), 3,000 duplicate pairs per rung on *one* shared seed list (prefix `bounded-v1`), so every budget replays identical deals and adjacent rungs compare per seed. The metric is mean set-share, $\text{sets}_A/(\text{sets}_A + \text{sets}_B)$, not win rate: the ladder needs a dial that keeps moving after the win rate saturates.

2. **E2 — tier calibration.** The three shipped tiers against the same reference opposition, on the head of the E1 seed list, so each tier's share sits on the E1 curve per-deal and its bits-equivalent is read off by interpolation.

3. **E3 — evidence age.** No games of its own and no engine instrumentation: computed post-hoc from the E1/E2 records, which retain the full public log. The estimator, fixed in the registration: at ask event i, every card whose public location a hit established at event j (unmoved and unretired since) that the acting seat could legally ask of the correct holder is one availability observation at age $i - j$; the seat *exploited* it iff its actual ask was exactly that card at that holder; the hit column is the actual ask's outcome. Hands are reconstructed from the seed's deal plus the log's hits and claims, so ask legality is decided from ground truth. Clustered comparisons cluster by seed; the young band is ages 1–8, the old band 33+; a half-life is the lower edge of the first band with at least 200 observations whose exploit rate is at or below half the youngest band's — a curve that never falls that far has none.

4. **E4 — style under memory pressure.** The v1.0 classifier-accuracy harness [5, 6], budget-capped: for each bits $\in \{16, 32, 64, \infty\}$, all 36 unordered style pairings play 50 single games with *both* teams bounded at that budget, on the v1.0 accuracy seed list (`clsacc-v1`); the classifier reads every seat once at end of game against the committed full-strength fingerprints [15] — 1,800 games and 10,800 read seats per cell. At ∞ the arm is decision-identical to the bare styles by the anchor pin, so that cell is a re-run of the v1.0 accuracy experiment and must reproduce the committed 22.4% exactly.

5. **E4b — single-seat attribution** (registered after the Phase 2 review; Section 6.4). E4's grid, seeds and pairings, but only one seat is bounded: the *read seat*, seat $2 \cdot (\text{game mod } 3)$ — the three team-0 seats in rotation — always playing the pairing's team-0 style under the budget while the other five seats play their bare full-strength styles exactly as the v1.0 harness seats them. The start seat rotates game mod 6, so the read seat occupies every relative table position uniformly. P8 scores the read seat only; truth is always the pairing's first style, so the triangular pairing scheme gives by-style read counts that fall with roster position, identically at every budget, and the last roster style is never the read truth. The ∞ cell carries an in-run health gate: every ∞-budget game is replayed with all six seats bare and required to match event-for-event, read-for-read and step-for-step (replays are checks, not records). The design holds the ecology — opponents, partners, game length — approximately at the full-strength distribution the fingerprints were calibrated on, isolating the read seat's own signature.

6. **E4b-power — the P8 run of record** (registered after the E4b review; Section 6.4). The same design at 300 seeds per pairing — 10,800 reads per cell, matching P7's read count — on a fresh seed prefix (`clsacc-power-v1`) disjoint from the pilot's; mapping, grid, estimator and rule unchanged.

6.3 Predictions and verdict rules

Eight predictions travel verbatim inside the artifact [17]; they are typeset here with mathematical symbols but otherwise as registered. P1–P7 were registered 2026-08-29, before any Phase 2 run; P8 on 2026-08-30, before any E4b run.

P1 E1 (ladder monotonicity): set-share is non-decreasing in bits, within 2·(paired SE) at every adjacent pair.

P2 E1 (health, not prediction): at ∞ the pairing is an exact mirror; duplicate-deal set-share is 0.5000 exactly by construction. A deviation is a harness bug, full stop.

P3 E2 (question, direction only): each shipped tier lands at a finite, orderable bits-equivalent (easy < medium < hard). No numeric prediction — this is the measurement the ladder exists to provide.

P4 E3: full-memory policies are age-flat (the S48 log-reader regime, by construction).

P5 E3: bounded policies decay with age, and the decay half-life increases with bits (the S45 signature, by construction of spotlight + recency eviction).

P6 E3: the old easy tier (`logWindow` 6 + 25% uniform noise) is age-flat inside its window and cliff-edged at it — i.e. noise is not human-shaped; the paper's motivating contrast.

P7 E4 (direction): top-1 accuracy is non-increasing as bits shrink — memory pressure erodes the behavioural signature. Where it lands relative to the 22.4% full-strength baseline is the measurement, not a prediction.

P8 E4b (single-seat attribution): top-1 accuracy at the bounded read seat is non-increasing as its bits shrink, by the same adjacent-rung 2·SE rule as P7 (any violating rung refutes). The ∞ cell must reproduce the corresponding full-strength read exactly (health, not prediction).

The verdict rules were fixed with the predictions and ship in the code that assembles the artifact [19], all on the adaptive suite's 2·SE discipline. P1, P7, P8: every adjacent rung must satisfy $\Delta \geq -2 \cdot \mathrm{SE}$ (per-seed paired); any violation refutes. P2: integer-exact mirroring of every ∞ pair; a deviation refutes and independently voids the run. P3: confirmed iff all three tiers are finite and strictly ordered; ordered-but-not-all-finite is mixed. P4: a full-memory policy is age-flat iff $|\mathrm{young} - \mathrm{old}| \leq 2 \cdot \mathrm{SE}$ (seed-clustered); both of the two full-memory policies flat confirms, one is mixed. P5: every finite budget must decay significantly *and* have a defined half-life, non-decreasing in bits with at least one strict increase; anything between "all hold" and "none decays" is mixed. P6: the old easy tier must be flat inside its window ($|\mathrm{rate}(1\text{–}3) - \mathrm{rate}(4\text{–}6)| \leq 2 \cdot \mathrm{SE}$) *and* cliff-edged at it; one of two is mixed. A refuted prediction is emitted as `refuted`, not massaged.

6.4 The registration timeline, and a correction of record

The sequence matters more than usual in this paper, because it includes a correction, and the correction is part of the method rather than a blemish on it.

1. **2026-08-29**: E1–E4 designs and P1–P7 registered; the base suite runs; P7 is refuted (Section 7.4).
2. **Phase 2 review**: the review confirms that E4's both-teams design *cannot* attribute the P7 refutation — bounding both teams changes the whole ecology (opponents, partners, game length) along with the read seat's signature, and the classifier's fingerprints were calibrated on full-strength play. The review also recommends a multiplicity annotation: Bonferroni ×3 over each adjacent-rung family, reported *alongside* the committed verdicts, which are not changed.
3. **2026-08-30, before any E4b run**: E4b registered — the single-seat design, P8, and the annotation; the read-seat mapping was chosen by the implementer and documented in the run's `meta.notes` before running, as the registration required.
4. **The pilot and its overclaim**: the 50-seed E4b pilot runs; P8 comes out confirmed (flat). The commit recording it claimed the P7 shift was "not of the read seat's signature". That claim was too strong: what a flat curve at 1,800 reads per cell licenses is only that the effect *does not appear in the single-seat design at that read count.*
5. **The E4b review**: computes the pilot's power at exactly the P7 effect size: about 52% (minimum detectable effect 0.0051 against the P7 effect 0.0053; the pilot's 64→∞ interval $[-0.0067, +0.0034]$ contains the entire P7 effect). The pilot's CONFIRMED is therefore an underpowered null.
6. **2026-08-30, after that review and before any run**: E4b-power registered. Seeds 50 → 300 (reads 1,800 → 10,800 per cell, matching P7's read count; MDE $\approx$0.0021 under the same rule), fresh disjoint seed prefix, everything else unchanged. The 300-seed run becomes the P8 verdict of record; the pilot is retained verbatim in the artifact, clearly labelled; both are

Table 1: E1, the ladder: three bounded-Balanced seats vs. an unbounded Balanced team at hard skill, 3,000 duplicate pairs per rung on one shared seed list [17]. Δ is the per-seed paired delta from the previous rung; the pre-registered rule is $\Delta \geq -2 \cdot \mathrm{SE}$ at every rung.

Bits	Set-share	SE	Mean moves	Δ from prev.	z
0	0.1313	0.0017	798.2	—	—
8	0.2449	0.0022	768.9	$+0.1135 \pm 0.0028$	41.11
16	0.3921	0.0026	741.1	$+0.1472 \pm 0.0031$	46.76
24	0.4361	0.0023	720.5	$+0.0440 \pm 0.0030$	14.50
32	0.4653	0.0020	704.5	$+0.0292 \pm 0.0025$	11.49
48	0.4909	0.0011	688.5	$+0.0257 \pm 0.0020$	12.52
64	0.4977	0.0006	685.1	$+0.0068 \pm 0.0011$	6.04
96	0.4999	0.0001	683.3	$+0.0022 \pm 0.0006$	3.80
128	0.5000	0	683.2	$+0.0001 \pm 0.0001$	0.90
∞	0.5000	0	683.2	$+0.0000 \pm 0.0000$	0.00

reported whatever they say. The registration notes the direction of the risk honestly: raising N after a null could only have refuted the project's own headline attribution — which is the direction honesty points. A correction commit states the pilot conclusion's within-design limit; no history is rewritten.

7. **Guard truthfulness**: the same review's integrity findings are fixed before the power run — the artifact-extension guards' prose narrowed to what they actually check, the registered prediction texts authenticated against the code's own copies, the base artifact pinned by records digest, and the analysis step made to fail (not note-and-proceed) when the records file is absent [19].

7 Results

Health first, because a run is void without it: across all three runs (base, pilot, power) the counters are clean — `illegalActions`, `cappedGames`, `invariantViolations`, `ties`, `voids` and `nonClinch` all zero, and `distinctSeeds` equal to the expected count in every cell [17].

7.1 P1 and P2: the ladder is monotone, and the anchor holds

Table 1 is the axis. Set-share rises from 0.1313 ± 0.0017 at zero bits to exactly 0.5 at the top, and every one of the nine adjacent deltas is non-negative — the first seven each individually significant (the smallest at $z = 3.80$), the two top rungs statistically zero. **P1 is confirmed**: 0 of 9 rungs violate $\Delta \geq -2 \cdot \mathrm{SE}$; the most negative rung is $128 \to \infty$ at $z = 0.00$. **P2 holds as a health gate**: all 3,000 ∞-budget pairs mirrored integer-exactly (set counts and step counts compared as integers, never floats), 0 deviations, duplicate-mean set-share 0.5000 — the run-scale echo of the Phase 1 equivalence anchor.

Two structural readings. The curve is concave with an elbow between 16 and 48 bits: the first 16 bits buy 0.26 of set-share, the next 32 buy 0.10, everything past 64 buys 0.0023. And the axis saturates early: at 128 bits every pair already mirrors exactly and the per-seed set counts match the ∞ cell's (Δ and SE both zero) — not because the games are identical (the two cells' mean lengths differ by about 0.002 moves), but because no retained-versus-evicted difference above 128 bits ever moved a set on these 3,000 deals. The worst-case pool is hundreds of bits (Section 4); the *operative* pool, under this eviction policy and these deals, fits in 128. Bounded games are also longer: mean

Table 2: E2, tier calibration: each shipped tier vs. the same reference opposition on the head of the ladder seed list, 3,000 pairs each, interpolated onto the E1 curve [17].

Tier	Set-share	SE	Bits-equivalent
easy	0.0446	0.0011	0 $[0, 0]$ — clamped: below the 0-bit floor
medium	0.4656	0.0024	32.2 $[30.8, 35.1]$
hard	0.5006	0.0007	none finite — above every finite rung

engine moves run from 683.2 at ∞ up to 798.2 at the floor, about 17% longer — memory-poor tables take longer to finish, a fact that returns in Section 7.4.

7.2 P3: where the shipped tiers land

Table 2 is the measurement the ladder exists to provide, and the verdict is **mixed**, exactly as the rule reads it: the ordering easy < medium < hard holds on point estimates, but not every tier is finitely placeable.

The medium tier is worth 32.2 bits $[30.8, 35.1]$ — a genuinely useful number: the full knowledge engine without constraint-refined hit probability plays like a contestability-ranked memory of about sixteen card facts. The hard tier's share, 0.5006 ± 0.0007, sits above every finite rung (its interval maps to $[85.5, \infty)$ in bits); it is not the mirror of the reference — the nonzero paired SE says the two disagree on some deals — but no finite budget reproduces it.

The easy tier is the headline of this table: 0.0446 ± 0.0011, *below the ladder's own zero-bit floor* of 0.1313 ± 0.0017. The bits-equivalent is clamped at zero with the artifact's note printed as measured: the share sits at or below the 0-bit floor. Read plainly: a bot that remembers *nothing* but reasons soundly over its own hand and the public board state, and never blunders, is worth nearly three times the set-share of a bot that remembers six events and dices away a quarter of its asks. The noise is not a memory model; it is a tax collected at random, and Section 7.3 shows its shape is wrong too.

7.3 P4–P6: the evidence-age curves

E3 asks the S45 question of every policy in the E1/E2 records: when a certain ask is available — a card whose holder the public log established at a known past event — does the policy take it, and does the answer depend on how old the evidence is? Table 3 reports the committed statistics per policy.

P5 — mixed, with the predicted structure where the model binds. Every budget from 0 through 48 bits decays significantly, and the half-lives, where defined, are 5/13/17/33/49 events at 0/8/16/24/32 bits — strictly increasing in bits, the S45 signature arriving exactly as the construction predicts: a bigger budget keeps old facts alive longer. But the registered rule demanded decay *and* a defined half-life at *every* finite budget, and from 48 bits up no band meeting the registered 200-observation floor falls to half the youngest-band rate (48 bits still decays at $z = 3.22$ but earns no half-life; 64–128 bits are statistically flat). The verdict prints as mixed: every budget decays — no; half-life defined everywhere and non-decreasing with a strict increase — no.

P4 — mixed, and instructive about estimators. The ∞-budget bounded policy is age-flat under the registered rule (-0.0145 ± 0.0129, $z = -1.13$ over 470 seeds). The unbounded reference is *not*: -0.0212 ± 0.0049, $z = -4.28$ over 2,571 seeds — a small but significant tilt in the *anti*-decay direction (within a seed, old-evidence certain asks are exploited slightly more often than young ones). One of two full-memory policies flat is, by the fixed rule, mixed. The instructive part is the

Table 3: E3, evidence age: exploit rate of available certain asks, young band (ages 1–8) and old band (33+), with the seed-clustered decay (young − old within seed) and the half-life where defined [17]. Pooled and clustered estimators can disagree (see text).

Policy	Young rate	Old rate	Clustered decay ± SE	z	Half-life
bounded 0	0.1030	0.0186	$+0.0661 \pm 0.0026$	25.22	5
bounded 8	0.2798	0.0574	$+0.1611 \pm 0.0052$	30.72	13
bounded 16	0.3626	0.1154	$+0.1055 \pm 0.0076$	13.92	17
bounded 24	0.4553	0.1646	$+0.0891 \pm 0.0104$	8.55	33
bounded 32	0.5060	0.2632	$+0.0572 \pm 0.0119$	4.81	49
bounded 48	0.5491	0.3626	$+0.0422 \pm 0.0131$	3.22	—
bounded 64	0.5576	0.4091	-0.0044 ± 0.0130	−0.34	—
bounded 96	0.5595	0.4244	-0.0145 ± 0.0129	−1.13	—
bounded 128	0.5595	0.4244	-0.0145 ± 0.0129	−1.13	—
bounded ∞	0.5595	0.4244	-0.0145 ± 0.0129	−1.13	—
reference (unbounded)	0.5840	0.4561	-0.0212 ± 0.0049	−4.28	—
old easy tier	0.4394	0.0266	$+0.4115 \pm 0.0024$	170.20	7

Table 4: E3, the old easy tier's exploit rate and hit rate by evidence age [17]. The memory window is 6 events; the cliff between the 5–6 and 7–8 bands is the window's edge, seen from outside.

Age band	Available	Exploit rate	Hit rate
1–2	34,816	0.5097	0.8015
3–4	46,943	0.4936	0.8067
5–6	25,064	0.5265	0.8084
7–8	17,138	0.0205	0.5557
9–12	30,822	0.0359	0.4717
13–16	24,370	0.0300	0.4728
17–24	37,003	0.0279	0.4702
25–32	25,595	0.0251	0.4703
33–48	30,450	0.0252	0.4611
49–64	14,319	0.0260	0.4548

composition effect the artifact makes visible: pooled, the reference's young and old rates are 0.5840 and 0.4561 — an apparent 13-point decay — while the seed-clustered estimator, comparing within games, finds the opposite sign. Both numbers are printed because they measure different things: games in which 33+-aged certain asks are still *available* at all are systematically unusual games (long, contested, with located cards left unconsumed), so the pooled old band conflates position quality with age. The clustered estimator is the verdict's input for exactly this reason, and wherever the two disagree in this paper, both are shown.

P6 — mixed, and the motivating contrast lands. Table 4 is what "25% noise plus a window" looks like as a memory model. Inside the six-event window the exploit rate holds near 0.51; at age seven it falls off a cliff to 0.02 and stays there for the rest of the game. The registered cliff statistic is $+0.4861 \pm 0.0019$, $z = 258.87$ — for scale, the largest cliff any bounded budget shows at the same boundary is 0.1215 ± 0.0033 (8 bits), and the unbounded reference's is 0.0718 ± 0.0027. But the tier is *not* flat inside its window: the ages 1–3 versus 4–6 split is -0.0360 ± 0.0035 ($z = -10.31$) — the rate *rises* slightly with age inside the window — so one of the two registered conditions fails and the verdict prints mixed. The substantive contrast survives the mixed verdict intact: human memory, and the bounded ladder built to imitate it, decays smoothly with half-lives that scale with capacity; the noise tier is a step function — perfect recall to a hard edge, near-total amnesia beyond

Table 5: E4: v1.0 classifier end-of-game top-1 over ecologies with both teams bounded, 36 pairings $\times$ 50 games $\times$ 6 seats per cell [17]. Chance is $1/9 \approx 0.111$; the committed full-strength baseline is 0.2243 [16]. Δ is the per-seed paired adjacent delta; rule $\Delta \geq -2 \cdot \mathrm{SE}$.

Bits	Seats	Top-1	Δ from prev.	z
16	10,800	0.1542	—	—
32	10,800	0.2101	$+0.0559 \pm 0.0073$	7.70
64	10,800	0.2295	$+0.0194 \pm 0.0050$	3.87
∞	10,800	0.2243	-0.0053 ± 0.0019	**−2.75**

Table 6: E4 per-style end-of-game top-1 by budget [17]. Descriptive — the artifact commits no per-style SEs; 1,200 seats per entry. The ∞ column is the v1.0 baseline's per-style accuracy, reproduced exactly.

True style	16 bits	32 bits	64 bits	∞
Balanced	0.0158	0.0433	0.0600	0.0558
Blitz	0.0592	0.0758	0.0925	0.0925
Punter	0.2742	0.1717	0.1333	0.1292
Banker	0.1075	0.1375	0.1250	0.1258
Turtle	0.2733	0.3525	0.3142	0.2717
Hoarder	0.1825	0.2575	0.3125	0.3092
Scout	0.2667	0.3675	0.4133	0.4125
Ghost	0.1608	0.3775	0.4983	0.5042
Archivist	0.0475	0.1075	0.1167	0.1175

it — and no human plays like that [9, 2].

7.4 P7 refuted: memory pressure makes styles *more* classifiable

P7 predicted that memory pressure erodes the behavioural signature: less memory, less characteristic play, lower classifier accuracy. Table 5 refutes it at both ends. At the bottom of the grid the direction is as predicted — 16-bit ecologies read at 15.4%, well below the 22.4% full-strength baseline. But the top of the grid inverts: *64-bit ecologies are more classifiable than full-strength ones*, 23.0% against 22.4%, and the registered rung test rejects — the $64 \to \infty$ delta is -0.0053 ± 0.0019, $z = -2.75$, past the $\Delta \geq -2 \cdot \mathrm{SE}$ bound. One violating rung refutes: **P7 is refuted**. The review's multiplicity annotation does not soften it: over the three-rung family, the violating rung's one-sided $p = 0.0029$ becomes $p = 0.00885$ under Bonferroni $\times 3$ — still a violation at the corrected level. The ∞ cell reproduces the committed v1.0 baseline to the last digit (0.22425925... in both artifacts) — the same games, by the anchor pin — so the comparison against v1.0 is an identity, not a replication.

Where does the extra classifiability come from? Table 6 (descriptive — no per-style SEs are committed) shows the readings redistribute rather than uniformly rise: under deep pressure the Ghost, v1.0's one recognisable style, collapses (0.5042 at ∞ to 0.1608 at 16 bits) while the Punter more than doubles (0.1292 to 0.2742) — bounded ecologies push behaviour into regions where different fingerprints are nearest. Two mechanisms plausibly combine, and E4 cannot separate them: the read seat's own behaviour changes, and the *ecology* changes — with all six seats bounded, opponents miss more, games run longer (Section 7.1 measured 17% on the one-sided ladder), and every length-sensitive feature shifts against fingerprints calibrated on full-strength play [15, 6]. Which of the two carries the P7 shift is precisely what E4 cannot say — the Phase 2 review's finding, and the reason E4b exists.

Table 7: E4b: single-seat top-1 at the bounded read seat, pilot (50 seeds, 1,800 reads per cell) and power run (300 seeds, 10,800 reads per cell; the P8 verdict of record) [17]. Cell SEs are seed-clustered; Δ per-seed paired; rule $\Delta \geq -2 \cdot \mathrm{SE}$.

	Pilot (1,800 reads)		Power run (10,800 reads)	
Bits	Top-1	Δ from prev.	Top-1	Δ from prev.
16	0.1678 ± 0.0100	—	0.1622 ± 0.0040	—
32	0.1678 ± 0.0104	$+0.0000 \pm 0.0118$	0.1698 ± 0.0044	$+0.0076 \pm 0.0046$
64	0.1700 ± 0.0105	$+0.0022 \pm 0.0066$	0.1642 ± 0.0043	-0.0056 ± 0.0031
∞	0.1683 ± 0.0102	-0.0017 ± 0.0026	0.1634 ± 0.0043	-0.0007 ± 0.0010

7.5 P8: the single-seat design, at pilot and at power

Table 7 carries both E4b runs, pilot and power, side by side — both reported whatever they say, as registered. In the power run — the P8 verdict of record — the single-seat curve is flat: 16.22%, 16.98%, 16.42%, 16.34% across the grid, no adjacent rung violating the rule (the most negative, 32→64 at -0.0056 ± 0.0031, $z = -1.83$, sits inside the 2·SE bound; its Bonferroni-corrected one-sided p is 0.1018). **P8 is confirmed**, now at 99.97% post-hoc power at the P7 effect size (MDE 0.0019 under the registered rule) rather than the pilot's 52.4% (MDE 0.0051). The ∞ health gate held in both runs: every ∞-budget game — 1,800 in the pilot, 10,800 in the power run — reproduced the all-bare v1.0 harness game event-for-event, read-for-read and step-for-step, 0 deviations.

Two features of the single-seat numbers deserve explicit note. First, the absolute level: a single bounded seat among five full-strength ones reads at about 16–17% at every budget. That level is not comparable to E4's cells or to the v1.0 baseline: P8's reads are always the pairing's *first* style, so the triangular pairing scheme weights the read distribution toward the early roster (2,400 Balanced reads down to 300 Ghost, and the Archivist is never the read truth), identically at every budget. The within-column comparisons P8 is built on are unaffected; the level is not the v1.0 baseline's population and is not read against it. Second, the cross-design comparison the E4b-power registration required: at the violated rung (64→∞), P7's delta is -0.0053 ± 0.0019 and P8's is -0.0007 ± 0.0010; the difference of deltas is $+0.0045 \pm 0.0021$, $z = 2.11$, two-sided $p = 0.035$ — computed with independent errors, the power run's seed list being disjoint from E4's by construction. It is *labelled cross-design and enters no verdict*: P7's rung lives in a world where both teams are bounded, P8's in a world where only the read seat is, and a difference between the two designs is exactly what the attribution story predicts rather than a within-design test of anything.

7.6 The attribution, stated at its licensed strength

Assembled: E4 measures that bounding *everyone* at 64 bits makes end-of-game style reads better than full strength, refuting P7. E4b measures that bounding *only the read seat* produces a flat curve, at a power (99.97%) that would almost certainly have detected the P7-sized effect had it been present in that design. The licensed conclusion: within the designs measured, the P7 shift travels with the ecology — opponents, partners, game length, the event mix the features are computed over — and not with the read seat's own end-of-game signature. What remains unlicensed, and is not claimed: that the read seat's signature is unaffected in E4's bounded-vs-bounded world (no design here isolates that), or any cross-design equality (the $z = 2.11$ difference is reported as a labelled observation). The pilot alone licensed none of this — its confidence interval contained the entire effect it was probing for — which is why the record distinguishes the pilot from the run of record, and why the correction of Section 6.4 was worth a commit.

Table 8: v1.5 after the defusal term: mean paired set-difference per duplicate pair against a *pre-concession* v1.0. Every row but the last is a *new* measurement made for this addendum with `scripts/probe-v15-clean.mjs` [18] — bank `v15clean-2026a`, the replication row on the disjoint bank `audit-2026q3-repl`, 2,000 duplicate pairs per cell, both zero-appetite controls exactly 0.0000 ± 0.0000 and every health counter zero over the 68,000 games each bank runs; neither BOUNDED.md nor CONCESSION.md carries them. Positive is v1.5 ahead. The last row is BOUNDED.md §5a's committed cell [13], which measured the *pre-concession* v1.5 against the same clean v1.0; it is printed beside them and is not superseded — it answers a different question.

Arm, at ∞ bits	Win rate	Paired set-difference	n
updated v1.5 (balanced) vs. clean v1.0	60.22%	$+1.4065 \pm 0.1399$	2,000
independent replication, disjoint bank	61.27%	$+1.5290 \pm 0.1418$	2,000
updated v1.5 at style punter vs. clean v1.0	62.10%	$+1.6595 \pm 0.1413$	2,000
updated v1.5 vs. *old* v1.5	—	$+1.5725 \pm 0.1405$	2,000
zero-appetite controls, both arms	—	0.0000 ± 0.0000	—
§5a, committed: old v1.5 vs. clean v1.0	—	-0.1255 ± 0.0590	2,000

8 Addendum: the ladder under the defusal term

Everything above was measured before the engine had a concession layer. It has one now [8, 14]: `defuse`, a term added to the ranked ask score that credits an ask for the reach a hit would strip off an opponent able to punish a conceded turn. All nine roster styles carry it at appetite 1; no shipped tier does. Its sibling `conceal` is 0 on all nine roster styles and on all three tiers, so the only live mechanism in the comparison below is **defusal**, and it is named that here rather than by the layer it arrived in. Because the bounded arm plays a *named roster style* at hard skill over restricted knowledge (Section 5), v1.5 inherited the term without a line of v1.5 changing. That inheritance is what makes a re-measurement necessary, and — as the second finding below shows — what makes the re-measurement easy to get wrong.

The comparison being re-run is BOUNDED.md §5a's [13], which had measured the then-current v1.5 against v1.0 on a held-out bank and found -0.1255 ± 0.0590 sets per duplicate pair at an unbounded budget. That residual is not a memory effect: at ∞ bits the v1.5 seat is byte-identical to a bare Balanced seat by the equivalence anchor (Section 5.1), so what it prices is the *style* gap against a v1.0 seat that spends 44.2% of its decisions as Punter [13]. Naming Punter reversed it there, and the ladder against a v1.0 opponent was monotone at every rung, as it is against v0.5.

Table 8 is the re-measurement, replicated on a disjoint bank and repeated at a second style. The updated arm beats the same clean v1.0 by $+1.4065 \pm 0.1399$ and $+1.5290 \pm 0.1418$ sets per pair; at style punter it takes $+1.6595 \pm 0.1413$; measured head-to-head against its own pre-concession self it takes $+1.5725 \pm 0.1405$. Both zero-appetite controls printed exactly 0.0000 ± 0.0000 — the arm with the term switched off is the arm without it — and no health counter fired across 68,000 games. Four things follow, and each corrects an overstatement the table invites.

First: §5a's committed cell is neither superseded nor wrong. It measured the *old* v1.5 against a clean v1.0; Table 8's top rows measure the *updated* v1.5 against the same clean v1.0. Different question, different answer, and the sign change between them is the term's arrival rather than a correction of anything. The provenance is checkable and was checked: an audit reproduced the artifact behind §5a per pair from a pre-concession export with **0 mismatches in 2,700 pairs**, and from the post-concession tree with **1,369 of 2,700**. That pair of counts, not any agreement of means, is what establishes the committed cell was measured on a clean tree.

Table 9: The same comparison across the budget ladder — v1.5 (balanced) minus a pre-concession v1.0, per duplicate pair, with 95% intervals — before and after the defusal term. The left column is BOUNDED.md §5a's committed ladder [13]; the right is new, measured for this addendum by `scripts/probe-v15-clean.mjs` on bank `v15clean-2026a`, 2,000 duplicate pairs per rung [18]. §5a's 8-, 24- and 96-bit rungs have no updated counterpart and are omitted; the two columns sit on different banks and are not paired against each other, so no row of this table measures the defusal term's own contribution.

Bits	§5a, pre-concession	Updated
0	-8.1450 ± 0.0689	-8.3920 ± 0.0645
16	-2.8520 ± 0.1497	-1.5635 ± 0.1656
32	-0.9940 ± 0.1200	$+0.5365 \pm 0.1518$
48	-0.2875 ± 0.0789	$+1.2200 \pm 0.1423$
64	-0.1625 ± 0.0641	$+1.3855 \pm 0.1400$
128	-0.1255 ± 0.0590	$+1.4065 \pm 0.1399$
∞	-0.1255 ± 0.0590	$+1.4065 \pm 0.1399$

Second, and this is the part worth carrying out of this repository: a contaminated run looks exactly like a faithful replication. Both generations resolve their style through the same `STYLE_ROSTER` — v1.0's `AdaptiveSpec` selects a roster entry per decision, v1.5's `BoundedSpec` names one — so an old-version arm driven by the current tree silently inherits the new mechanism, on both sides of the comparison at once. Running both arms post-concession gives -0.0740 ± 0.0533, statistically indistinguishable from §5a's committed -0.1255 ± 0.0590. A reviewer holding the two side by side would call the second a clean replication of the first. *Magnitude cannot detect this class of error; only provenance can*, which is why the audit above is a per-pair identity check against a pinned export rather than a comparison of intervals.

Third: the ladder's shape did not survive unchanged, and an early reading of these runs that said "only the level moves" is wrong. Table 9 puts the two columns side by side. The $0 \to \infty$ span grows from 8.0195 to 9.7985 on the first updated bank and 9.9260 on the second. The share of that span bought by the first 16 bits moves too, though far less: from §5a's own rungs, $(8.1450 - 2.8520)/8.0195 = 66.0\%$; on the first updated bank $(8.3920 - 1.5635)/9.7985 = 69.7\%$ and on the second $(8.3970 - 1.4730)/9.9260 = 69.8\%$. (§5a's prose puts the first of these at 59% and adds "73% by 8"; neither figure follows from the table printed beside it — 8 bits buys $(8.1450 - 6.0215)/8.0195 = 26.5\%$, so the pair is not only mis-stated but ordered backwards, a cumulative share falling as the budget grows. The recomputation above is used here in preference to either.) What moves the shape is that the term's own contribution is budget-dependent, and the contrast that isolates it is the concession-only one — updated v1.5 against *old* v1.5 at a matched budget, which holds the opponent fixed and leaves the term as the only difference between the arms. At zero bits it pays $+0.9560 \pm 0.1858$ on `v15clean-2026a` and $+0.8030 \pm 0.1879$ on the disjoint `audit-2026q3-repl`, 2,000 duplicate pairs each, against $+1.5725 \pm 0.1405$ and $+1.6380 \pm 0.1440$ at ∞ on the same two banks [18]. A term whose worth depends on the budget it sits under cannot leave that budget's curve alone, and it did not. Table 9 is not evidence for that budget-dependence and must not be quoted as if it were: its two columns face a *full-memory* v1.0 while the seat itself is starved, they sit on different banks with no pairing between them, and at zero bits both are near the floor, so their $-8.3920 - (-8.1450) = -0.2470$ difference is not the term's contribution and does not even carry its sign. One of §5a's caveats travels with the comparison just made, because that comparison is exactly a cross-budget reading of shape against §5a's baseline: the v1.0 baseline is not a fixed policy across rungs. Its style mixture tracks public-log length, which the opponent's budget changes — warm-Punter share runs 41.0% at ∞ to 48.6% at 0 bits — so cells at different budgets face materially different opponents [13].

Fourth, and this one qualifies the central axis of this paper. The defusal term's licence scan reads the *full public log* for row-6 asks and retires each licence only against the seat's budgeted knowledge. The retirement half is therefore priced; the ask history it scans is not. `decide.ts` names the asymmetry in its own comment as "the one thing the v1.5 cost model does not yet price", and CONCESSION.md §9 carries it as open follow-up — moving the licence into the fact pool as a first-class 1-bit basis read is the correct end state, and doing it now would change every committed v1.5 number [19, 14]. The consequence has to be drawn rather than left implicit: **"updated v1.5 at N bits" is not a clean N-bit agent**, and Table 9's right-hand column must not be read as a pure memory ladder. The concession-only contrast at zero bits is the proof of it — a seat that by construction retains nothing still collects $+0.9560 \pm 0.1858$ and $+0.8030 \pm 0.1879$ sets over its own pre-concession self, from a mechanism reading a log the budget was supposed to gate [18]. None of this touches Sections 7.1–7.6: those were measured before the term existed, and every number in them stands as printed.

9 Discussion

9.1 What the ladder buys

The replacement knob does what the old one could not, and the numbers are now attached. *Monotone*: nine of nine adjacent rungs non-negative under a paired test (Table 1). *Interpretable*: a tier is now a number — medium *is* 32.2 bits $[30.8, 35.1]$ — and a handicap can be stated in units a player can reason about. *Human-shaped*: in the regime where the budget binds (0–32 bits), effectiveness decays with evidence age at half-lives 5–49 events that rise with capacity (Table 3) — the S45 curve, produced by construction rather than tuned. *Composable*: E4 ran all nine styles under budgets without any per-style work, because the arm restricts the knowledge, not the policy.

The concavity of the curve is itself a finding about `us54`: the first 16 bits — eight card facts, or a few facts plus a clutch of basis bits — carry about 71% of the distance from amnesia to full strength (0.26 of the 0.37 set-share gap), and everything past 64 bits is worth under a third of a point of set-share. Contestability-ranked memory is extraordinarily efficient at the bottom: what matters in this game is a small, current working set, not the archive.

9.2 The noise tier, priced and shaped

The old easy tier now has two independent condemnations. Priced on the ladder, it is worth less than zero bits — the memoryless-but-sound floor beats it by nearly a factor of three in set-share (Table 2), because a 25% uniform blunder rate and a guessing declarer destroy more value than a six-event memory saves. And shaped on the evidence-age axis, it is a step function with a $z = 259$ cliff at its window edge (Table 4) where humans — and the bounded ladder — show smooth decay. A learner watching the noise tier sees perfect play interleaved with inexplicable blunders; a learner watching a 16-bit bot sees something recognisable: a player who follows the current fight closely and forgets what happened twenty asks ago. The replacement is not merely stronger-per-unit-of-difficulty; it fails the way the thing it simulates fails.

9.3 What P7's refutation means, and does not mean

The refutation is genuinely informative about the observation problem. The v1.0 instrument [6] was calibrated on full-strength play, and its accuracy was bounded by how rarely styles diverge on identical views [7, 11]. Bounded ecologies move play into regions the fingerprints never saw — longer games, more misses, different feature distributions — and the reads *redistribute*: the classifier is not recovering more style signal at 64 bits so much as reading a shifted population against fixed references, and the shift happens to raise top-1. That interpretation is consistent with the E4b result: hold the population at full strength except for one seat, and the effect vanishes at high power. It is also a caution for any deployed read of handicapped play: a classifier calibrated at full strength does not degrade gracefully under ecology shift — it changes its answers in structured ways (the Ghost collapse and Punter surge of Table 6) that look like signal.

What the refutation does not support: any claim that memory pressure leaves individual signatures intact in general. P8's flatness is a statement at end-of-game reads, over this roster, against these fingerprints, in a design where five of six seats hold the ecology fixed. The v1.5 measurements say nothing about mid-game reads, cross-budget fingerprints, or classifiers recalibrated on bounded play — all of which are the natural next instruments if bounded opponents ship at the play table.

10 Threats to validity

1. **The ladder is an operationalisation, not a canon.** "Bits" are the framework's accounting units under one tariff, one retirement rule, one subsumption rule, and one eviction policy; they are not Shannon information (a one-bit basis fact can carry more decision value than a two-bit card fact — the tariff prices storage, not worth). A different defensible spotlight scoring or tie order would produce a different — presumably similar, but unmeasured — curve. Bits-equivalents (Table 2) are relative to *this* ladder.

2. **E4's both-teams design, and what E4b licenses.** P7's refutation is a property of bounded-vs-bounded ecologies read by a full-strength-calibrated classifier; E4b isolates the read seat only within its own single-seat design, at end-of-game reads, at its measured power. The cross-design difference ($z = 2.11$) is an observation, not a test; no design here measures the read seat's signature *inside* a bounded ecology.

3. **Fingerprints are calibrated on full-strength play.** Every E4/E4b read uses the committed v1.0 fingerprints [15]. The P7 movement therefore conflates behavioural change with calibration mismatch by construction; that conflation is the finding's substance in Section 9.3, but it caps what any accuracy number here says about intrinsic distinguishability.

4. **The E3 estimator is itself an operationalisation.** "Certain asks" are defined through public hits only; availability is conditional on the position, and the composition effect of Section 7.3 shows how strongly availability selects on game character — hence the seed-clustered estimator for every verdict, with pooled rates printed beside it. Half-lives require 200-observation bands, so their absence at 48+ bits partly reflects old-evidence availability thinning (strong players consume certain asks young), not only flatness.

5. **Retained logs elide two derivable payloads.** The E1/E2 records keep the full public log event-for-event, but a claim's assignments and true holders, and the final score snapshot, are elided as derivable or unused by the E3 estimator [17]. A future estimator needing them would need a re-run, not a re-read.

6. **Single machine, single implementation, closed population.** One engine, one rule set (pinned by the artifact's rules hash), self-play within one roster; the usual transfer caveats [5] apply in full. The equivalence anchors and mirror gates certify harness symmetry and the bounded arm's consistency with the unbounded engine — not the engine's own correctness, which rests on its audits [4, 19].

7. **The addendum's ladder is not a clean memory ladder.** Section 8's updated rungs carry a term whose licence scan reads the full public log, so the budget caps only part of what that seat knows; those numbers describe the v1.5 arm as it now ships, not an N-bit agent. Its two columns also sit on different banks and are not paired against each other, and the run was exploratory — no verdict rule was fixed in advance of it. Everything in Sections 7.1–7.6 predates the term and is unaffected.

8. **The E4 grid is coarse.** Four budgets, and the P7 violation is one rung; the shape between 64 bits and ∞ is uninstrumented. The Bonferroni annotation addresses the family actually tested, not the grid that might have been.

11 Conclusion

FishAI v1.5 set out to replace a difficulty knob that lied — noise dressed as weakness — with one that tells the truth, and the measurements say it does: a bit-budget memory with contestability eviction is monotone at every measured rung, saturates by 128 bits, prices the shipped tiers (medium = 32.2 bits; the old easy tier below the zero-bit floor it was meant to sit above), and fails the way people fail, with decay half-lives that grow with capacity. The pre-registered suite kept the project honest twice over: P7's refutation — mild memory pressure makes whole ecologies *more* classifiable, not less — survived its multiplicity annotation and became the paper's most interesting result; and the E4b pilot's overclaim was caught by its own review, corrected on the record, and settled by a 300-seed run of record at matched read count, which localised the effect, within the designs measured, in the ecology rather than the seat. The correction sequence is reported in full because it is the method working as designed: registrations before runs, reviews after them, and conclusions cut back to what the data licence — whichever direction that points. A later measurement is reported in the same spirit (Section 8): after the engine gained a defusal term the arm inherits, the cross-generation comparison was re-run, and it neither supersedes the committed cell it resembles nor leaves the ladder's shape alone — and the term partly escapes the very budget this paper's axis is built on, which is printed as a limit on the new numbers rather than worked around.

Reproducibility. The engine is deterministic; the committed artifact [17] carries the configuration, seed prefixes, records digests, engine commits, rules hash, the verbatim predictions and all verdicts, with the E4b pilot retained beside the run of record; every number here is a field of it or of the committed v1.0 artifact [16], and the suite re-runs from `npm run bounded` [19].

12 Addendum: the turn-pass correction, and what it did not move

September 2026. `RULES_US54.md` §4 carried a rule marked `[DERIVED]`, and the inference was wrong. When a declare emptied the current turn-holder, the engine advanced the turn to the next seat in ascending cyclic order that still held cards — which, because teams alternate by seat parity, is seat $turn + 1$, an **opponent**, whenever that opponent holds a card. The owner's rulebook gives the turn to a **teammate**, exactly as row 20 already did when a seat's *own* declare emptied it. Measured over

Table 10: The bit ladder under the corrected rule and under the rule it replaces, on identical seeds. The right-hand column is the correction's whole effect on this axis.

bits	corrected	previous rule	delta	rung SE
0	0.0953	0.0953	0.0000	0.0015
8	0.2184	0.2187	−0.0003	0.0021
16	0.3873	0.3876	−0.0003	0.0026
24	0.4388	0.4389	−0.0001	0.0024
32	0.4670	0.4671	−0.0001	0.0019
48	0.4931	0.4933	−0.0002	0.0010
64	0.4978	0.4978	0.0000	0.0005
96	0.4999	0.4999	0.0000	0.0001
128	0.5000	0.5000	0.0000	0.0000
∞	0.5000	0.5000	0.0000	0.0000

Table 11: The ladder's ends and the shipped tiers, as published and as regenerated. Every move below is attributable to the concession layer by Table 10's isolation, which prices the rule correction at ≤ 0.0003.

	as published	regenerated
zero-bit floor	0.1313	0.0953
top of the ladder	0.5000	0.5000
easy tier	0.0446 — below the zero-bit floor	0.0337 — below it still, clamped to 0.0 bits
medium tier	32.2 bits [30.8, 35.1]	**18.0** bits [17.2, 18.7]
hard tier	—	23.3 bits [22.6, 24.0]

400 games and 3,079 declares, 56 out-of-turn declares emptied the turn-holder, 37 of those had a teammate still holding cards, and **19 gave the turn to the wrong team** — about one game in twenty-one.

The engine now routes both cases into the same `awaitPass`, the rules document is corrected, and its hash moved, so every artifact in this project was regenerated. This section reports what that did to the ladder, and the answer is: essentially nothing.

12.1 The isolation

The ladder was re-run twice on one tree, with the turn-pass rule as the only difference — same seeds, same budgets, same roster, same 3,000 duplicate pairs per rung.

The largest move at any rung is 0.0003, and every move is inside that rung's own standard error. A real rules defect, firing in about one game in twenty-one, is invisible to this instrument. That is worth stating as a finding rather than a relief: the ladder measures a difference between two seats that share the defect, and a symmetric perturbation cancels in a paired duplicate-deal design almost exactly as the design promises.

12.2 What did move, and it is not memory

The regenerated artifact nonetheless differs substantially from the one Sections 7.1 onward report, and **the difference is the concession layer, not the rule and not the budget.** The committed artifact those sections read was generated on 2026-08-30, and the concession layer [14] landed later the same day; every roster style now carries `defuse` = 1, and the bounded arm delegates to the roster.

Two readings follow and only the first is safe.

Safe: the paper's claims about the *axis* are unchanged. P1 is CONFIRMED again — set-share is non-decreasing at all nine adjacent rungs, and the most negative rung sits at z 0.00. P2 is CONFIRMED, with 0 integer-exact mirror deviations over 3,000 ∞-budget pairs. P3 is CONFIRMED, with the tier ordering easy $<$ medium $<$ hard intact. P7 is CONFIRMED. P4, P5 and P6 come back MIXED exactly as before. **Every verdict in this paper survives the regeneration**, and so does the headline refutation: memory pressure still makes styles easier to classify, with end-of-game top-1 rising 16.3% $\rightarrow$ 20.5% $\rightarrow$ 22.1% across 16, 32 and 64 bits against 22.0% at full memory.

Not safe: the shipped medium tier is no longer worth 32 bits. It prices at 18.0 bits now, and the honest reading is that *the tier did not get cheaper — the ladder got steeper*, because every rung of it gained a mechanism the tier does not carry. The three shipped tiers still carry `defuse` $= 0$ while every ladder rung now delegates to a roster style that carries `defuse` $= 1$, so the tier is being priced against a stronger ruler than the one Section 7.2 used. **A tier price is a statement about two things at once, and this addendum is the demonstration.** Re-running the ladder with `defuse` $= 0$ throughout would restore the comparison the shipped tiers deserve; it has not been run, and until it is, the 32.2-bit figure and the 18.0-bit figure answer different questions rather than disagreeing about one.

12.3 One cross-design number weakened

Section 7.5's difference-of-deltas between the whole-ecology and read-seat-only designs is $+0.0012 \pm 0.0020$ on the regenerated artifact (z 0.60, two-sided p 0.55), where the committed run read $+0.0045 \pm 0.0021$ at z 2.11. It was reported there as a cross-design comparison entering no verdict rule, and it enters none here either; what changes is that it no longer excludes zero. Nothing in this paper rested on it, which is why it was labelled that way in the first place.

References

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[2] J. R. Anderson and L. J. Schooler. Reflections of the environment in memory. *Psychological Science*, 2(6):396–408, 1991.

[3] S. Kannan. Literature game framework: specification and reference implementation (`wh_impl.txt`; `wh_lit.py`, the `ActivePlayer` memory accounting). Read in full by the project's research synthesis [9] (entry S44), which quotes the bit tariff verbatim. The framework's deck is the 52-card, seven-card-major variant; the memory model is independent of that choice, its constants are not.

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[8] A. Hsieh. FishAI v2.0: The concession layer, or why avoiding a dangerous opponent loses. FishAI project; companion paper, 2026.

[9] FishAI project. PLAYSTYLES.md — Literature / Canadian Fish: play styles, rule dialects, and engine specification. Internal research synthesis, 2026. Entries S44 (bounded memory; the Kannan framework and tariff), S45 (recency; the decay signature), S46 (own-hand-only), S47 (triage memorisation; the contestability eviction policy), S48 (the log-reader regime; observation vs. display).

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[12] FishAI project. BOT_LAB.md — play-style spectrum: research, experimental design, and metrics. Repository document; §4.3 health gates, §5 experimental design, 2026.

[13] FishAI project. BOUNDED.md — FishAI v1.5: the bounded-memory ladder, and what it prices. Repository document; §4 the experiment suite this paper reports, §5a the committed v1.5-against-v1.0 comparison and its caveats, 2026.

[14] FishAI project. CONCESSION.md — FishAI v2.0: the three-sided ask. Repository document; §3 the defusal term and its measurement, §5a concealment, §8a the detection floor, §9 the open follow-ups — including the bounded arm's licence, which is derived rather than retained, 2026.

[15] FishAI project. Committed fingerprint calibration `lib/engine/bots/data/fingerprints.ts`. 150 `us54` mirror games per style at full strength, seed prefix `fingerprint-v1`; generated 2026-08-24.

[16] FishAI project. Adaptive artifact `src/lab/data/adaptive-results.json`. 125,600 games; records digest `ce3316641a8f38fe`; the committed end-of-game classifier baseline (top-1 0.22425925 . . .) cited by the bounded artifact's baseline block; generated 2026-08-24.

[17] FishAI project. Bounded artifact `src/lab/data/bounded-results.json` (schema 3). Base suite: 85,200 games, records digest `fe829b581f665c9a`, engine commit `ab779f3` (recorded `ab779f3-dirty`); E4b pilot: 7,200 games, digest `52bfe90b52ea7029`; E4b power run: 43,200 games, digest `53668ad79186621d`; rules hash `b4c7fef7...` shared with v0.5/v1.0; generated 2026-08-30.

[18] FishAI project. Two-tree defusal re-measurement `scripts/probe-v15-clean.mjs`. Post-concession v1.5 driven against a pre-concession v1.0 export from a second module graph; `us54` duplicate deals, 17 cells $\times$ 2,000 pairs = 68,000 games per bank. Bank `v15clean-2026a` and the disjoint replication bank `audit-2026q3-repl`; both zero-appetite controls exactly 0.0000 ± 0.0000, every health counter zero; run 2026-08-31. These measurements are new to this addendum and are carried by neither BOUNDED.md nor CONCESSION.md.

[19] FishAI project. FishAI repository: engine, bots, simulation laboratory, and results site. Bounded arm: `lib/engine/bots/bounded.ts` (and `tests/bots/bounded.test.ts`, the equivalence anchor); experiment suite: `lib/lab/bounded.ts`, artifact boundary `src/lab/bounded-artifact.ts`, runner `scripts/bounded-sim.mjs`, 2026.